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Preamble

The present document provides some information about the HDL code available in the 03 Bitbucket directory. The code may not be optimal or elegant but it passes all the tests supplied in the Nand2Tetris project files.

`Bit` chip

Useful sources : project documentation and the *Sequential chips* (A2.4) section from the *Appendix 2: Hardware Description Language* document. Here is the code I came up with :

```
Mux(a=dffOut, b=in, sel=load, out=muxOut);
DFF(in=muxOut, out=dffOut);
Or(a=dffOut, b=false, out=out);
```

Looking at it as a classical software developer without the background of the documentation, one can find strange that the code runs fine despite the fact that the `dffOut` pin is not initialized !
The code also contains comments describing the logic followed at the time it was written.

`Register` chip

As stated in the documentation, this chip can be directly implemented from the `Bit` chip :

```
Bit(in=in[0], load=load, out=out[0]);
Bit(in=in[1], load=load, out=out[1]);
(…)
Bit(in=in[15], load=load, out=out[15]);
```

`RAM1` chip

A RAM chip with 1 register :

```
Register(in=in, load=load, out=out);
```

The `RAM1`, `RAM2`, and `RAM8` chips are simple extensions of the chip so nothing special to add here.

`RAM64` chip

Before taking a look at `RAM64.hdl` (written more than 3 years ago), I was expecting 10 lines of instructions or so. But, obviously, I struggled a lot to set the 8 load bits and the address part ! So, the code is certainly not optimized. The various comments have been translated into english and the code has been tested again.

`PC` chip

See the comments in the HDL file.

RAM512, RAM4K and RAM16K chips

All those chips were built on the same logic as for RAM64.